Home Assignment - 3

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① $f(x) = \begin{cases} kx^2 & 0 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$

(i) find the value of k

property of probability Density function

$$\int_0^2 kx^2 dx = 1$$

$$k\left[\frac{x^3}{3}\right]_0^2 = 1$$

$$k\left(\frac{8}{3} - 0\right) = 1$$

$$\boxed{k = \frac{3}{8}}$$

(ii) obtain the probability that the lecture each within 1 min of the end of the hour

Sol: $P(0 \le x \le 1) = \int_0^1 \frac{3}{8} x^2 dx$

$$= \frac{3}{8}\left(\frac{x^3}{3}\right)_0^1$$

$$= \frac{3}{8}\left(\frac{1}{3}\right)$$

$$= 1/8$$

②

X	0	1	2	3	4
P(x)	0.1	0.3	0.3	?	0.4

Sol: a) find $P(x=3)$

Total probability : 1

$$P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4) = 1$$

$0.1 + 0.3 + 0.3 + x + 0.1 = 1$

$0.8 + x = 1$

$x = 1 - 0.8$

$\boxed{x = 0.2}$

$P(x = 3) = 0.2$

(b) $P(x \geq 2) = P(x=2) + P(x=3) + P(x=4)$

$= 0.3 + 0.2 + 0.1$

$= 0.6$

③ (a) find k

$f(x) = k/2^x$

$\Sigma f(x) = 1$

$k\left(\frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4}\right) = 1$

$k\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}\right) = 1$

$k\left(1 + \frac{15}{16}\right) = 1$

$k\left(\frac{31}{16}\right) = 1 \Rightarrow \boxed{k = \frac{16}{31}}$

(b) find the cummulative probability distribution $F(x)$

$F(x) = P(X \leq x)$

value:

$f(0) = \frac{16}{31}$, $f(1) = \frac{16}{31} + \frac{8}{31} = \frac{24}{31}$

$f(2) = \frac{24}{31} + \frac{4}{31} = \frac{28}{31}$

$f(3) = \frac{30}{31}$, $f(4) = 1$

④ Given that

$$f(x) = \begin{cases} k\sqrt{x} & , 0 < x < 1 \\ 0 & , \text{exclusive} \end{cases}$$

(a) Evaluate k

$$\int_0^1 k\sqrt{x}\, dx = 1$$

$$= k\left(\frac{2}{3}\left[x^{3/2}\right]_0^1\right) = 1$$

$$\Rightarrow k\left(\frac{2}{3}\right)\left((1)^{3/2} - 0\right) = 1$$

$$\boxed{k = \frac{3}{2}}$$

(b) Evaluate $P(0.3 < x < 0.6)$ using density function.

$$= \int_{0.3}^{0.6} \frac{3}{2}\sqrt{x}\, dx$$

$$= \frac{3}{2}\left(\frac{2}{3} x^{3/2}\right]_{0.3}^{0.6}$$

$$= (0.6)^{3/2} - (0.3)^{3/2}$$

$$= 0.464 - 0.164$$

$$= 0.30$$

⑤ Given $n = 20$,

$P = 10\% = 0.1$, $q = 0.9$

$$P(X = k) = {}^nC_k\, p^k q^{n-k}$$

(a) no defective $(k=0)$

$${}^{20}C_0\, (0.1)^0 (0.9)^{20}$$

$= (0.9)^{20}$

≈ 0.121

(b) Plexacity 2 defectives.

$20C_2 (0.1)^2 (0.9)^{18}$

$= 190 \cdot 001 \times 0.180$

≈ 0.285

(c) Between 1 and 4

$P(1) + P(2) + P(3) + P(4)$

$= 0.270 + 0.285 + 0.190 + 0.090$

≈ 0.835

(6) 75% with state

then out of state $= 25$

q 95% $= 0.75$, $P = 25\% = 0.25$ out of state

$n = 9$

let X = number of out of state vehicles

we need

$P(X<4) = P(0) + P(1) + P(2) + P(3)$

$P(X=k) = 9C_k (0.25)^k (0.75)$

$P(0) = (0.75)^9 \approx 0.075$

$P(1) = 9 \times 0.25 \times (0.75)^8 \approx 0.225$

$P(2) = 0.300$

$P(3) = 0.2813$

final:

$P(X<4) = 0.075 + 0.225 + 0.300 + 0.233$

$= 0.833$

(7) (a) Find the mean and variance of the number of patients who recovered from the corona

Sol: $n = 15$

$p = 0.4, \quad q = 0.6$

(a) mean and variance

Mean $= np = 15 \times 0.4 = 6$

Variance $= \mu = npq = 15 \times 0.4 \times 0.6 = 3.6$

(b) Exactly 5

$P(x=5) = {}^{15}C_5 (0.4)^5 (0.6)^{10}$

≈ 0.186

(c) At least 10

$P(X \geq 10) = P(10) + \dots + P(15)$

$= 0.033$

(d) Between 3 & 8

$P(3 \leq X \leq 8)$

$= 0.76$

(e) At most 3

$P(X \leq 3) = P(0) + P(1) + P(2) + P(3)$

$= 0.30$

(f) None survive

$P(0) = (0.6)^{15} = \approx 0.00062$

(8) Given

probability of 2 diamondy in one trail small → poision

$\lambda = np \Rightarrow P(\text{diamond}) = \frac{13}{52} = \frac{1}{4}$, prob of 2 diamondy

$= \frac{1}{4} \times \frac{12}{51}$

$= 0.07$

$$P(x=k) = \frac{e^{-\lambda}\lambda^{t}}{t!} \quad (\lambda = 1.5)$$

$$= \frac{e^{-1.5}(1.5)^0}{0!} + \frac{e^{-1.5}(1.5)^1}{1!} + \frac{e^{-1.5}(1.5)^2}{2!}$$

$= 0.19$

(b) If 2% of length bulbs are defective

find (i) at least one is defective

$n = 100$

$P = 0.02$

$\lambda = np = 2$

(i) $P(x \geq 1) = 1 - P(0)$

$= 1 - e^{-2}$

$= 1 - 0.135$

$= 0.865$

(ii) Exactly 7

$$P(7) = \frac{e^{-2}2^{7}}{7!}$$

$= 0.0034$

(iii) $P(1 < x < 8)$

$P(2) + P(3) + \dots P(7)$

$= 0.86$

(9) (a) more than 2150 hours

mean $\mu = 2040$

SD $\sigma = 60$

Total bulbs $= 2000$

$$z = \frac{2150 - 2040}{60}$$

$= 1.83$

$P(z > 1.83) = 0.0336$

$= 2000 \times 0.0336$

≈ 67 bulbs

(b) less than 1950

$z = -1.5$

$P = 0.0668$

$= 2000 \times 0.0668$

≈ 134 bulbs

(c) between 1920 and 2160

z values

$1920 \rightarrow 2$

$2160 \rightarrow -12$

$P = 0.954$

$= 2000 \times 0.954$

≈ 1903 bulbs

(d) Exactly 1960

in continous distribution → Probability = 0

(10) Given,

failed (<40 %) = 10 %

Distinction (>75%) = 5 %

from z-table

- 10% → z ≈ 1.28

75 % → z = +1.64

we need

second division (50% - 60%)

using symmetry and z-table

P(50% - 60%) = 0.24